Mira · Capstone Submission Guide

AI Project Intelligence Assistant for Nexora · Interview Kickstart, Applied Agentic AI for PMs/TPMs · Ashutosh Lonkar · November 2026

Mira is a multi-agent n8n workflow that helps Nexora's PMs and TPMs run the ABCDE Ltd. AI Adoption Project. It generates project plans, risk assessments, weekly status reports, milestone alerts and stakeholder updates from the project's own files, and asks for missing information instead of guessing. Every request is traced in Langfuse. Reports are saved to Google Drive as formatted documents with one-click PDF and Word downloads, and a scheduled workflow emails the weekly status report.

Design principle: code computes, the LLM writes. Every count, date comparison and filter is computed in a JavaScript node; each agent receives only the facts it needs and is told to copy them exactly.

How to read the tags: **(Core requirement)** = required by the brief. **(Extended capability)** = one of the brief's optional capabilities. **(Enhancement)** = added beyond the brief.

Results at a glance

Metric	Result	Source
Baseline tests passed	12 / 12	Langfuse annotation (test_pass)
Requests routed to the intended agent	12 / 12	Test log
Correct refusals on vague input (T2, T4, T6, T9)	4 / 4	Test log
Counts and task IDs matching the source files	100%	Checked against the CSVs
Average PM usefulness	4.5 / 5	Langfuse annotation (pm_usefulness)
Average latency per request	10.6 s	Langfuse traces
Average cost per request	$0.00046	Langfuse traces (estimated tokens)
Projected monthly cost (10 PMs × 8 requests/day × 21 days)	$0.78	Test log projection
Extended capabilities delivered	2 (Milestone Alerts, Stakeholder Updates)	Workflow

Cost uses estimated token counts (characters ÷ 4): the order of magnitude is reliable, exact cents are not.

What Mira does

Agents and capabilities

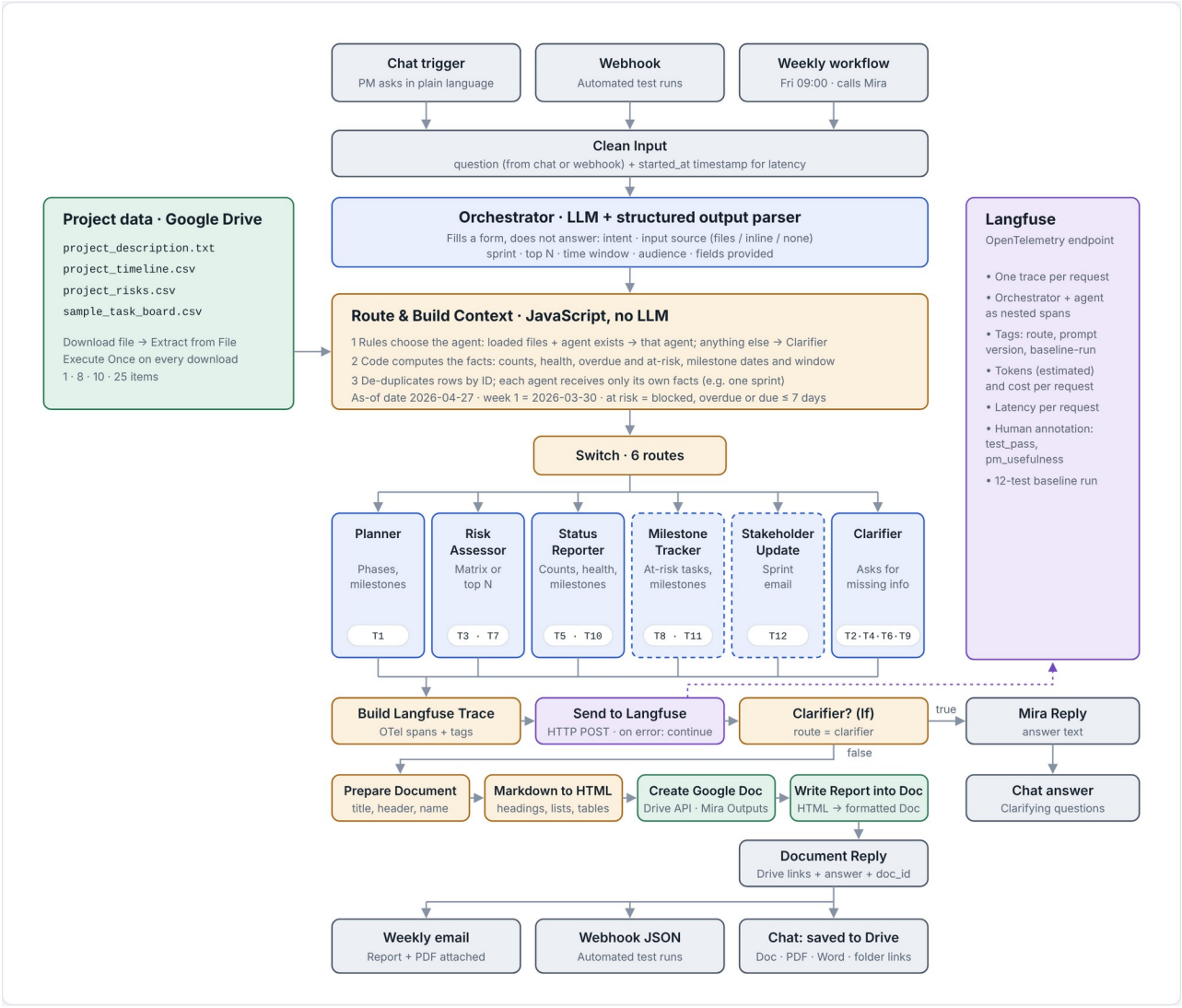
Capability	What it does	Tests
Project Plan Generation (Core requirement)	Planner Agent: 8 phases with weeks, activities, 14 milestones, goals copied from the description; open questions instead of invented dates, budget or team size	T1
Risk Assessment (Core requirement)	Risk Assessor Agent: full risk matrix (all 10 risks) or top N, each with category, impact, mitigation and reasoning	T3, T7
Weekly Status Report (Core requirement)	Status Reporter Agent: health (Red/Amber/Green rule), counts, tasks by status, needs attention, upcoming milestones; scoped to a sprint or the whole project	T5, T10
Upcoming milestones in the status report (Enhancement)	Added after the baseline review: current phase, milestones due in the next 14 days and next phase, computed from the timeline	T5
Milestone Alert System (Extended capability)	Milestone Tracker Agent: blocked and at-risk tasks with reasons and suggested actions; milestones in a time window	T8, T11
Stakeholder Update Generator (Extended capability)	Stakeholder Update Agent: sprint update email for leadership, engineering or client; only that sprint's tasks reach the agent	T12

Capability	What it does	Tests
Clarifier for vague requests (Enhancement)	Clarifier Agent: says the input is insufficient, asks specific questions, never generates a plan, risk list or status	T2, T4, T6, T9

Platform and design

Component	What it does
Orchestrator + Router pattern (Core requirement)	LLM with a structured output parser fills a form (intent, input source, sprint, top N, window, audience); rules in code choose the agent
Data grounding from project files (Core requirement)	Four Google Drive files read on every request (Download file → Extract from File)
Analytics layer in code (Enhancement)	Route & Build Context computes counts, health, overdue and at-risk tasks, timeline dates and milestone windows before any agent runs
De-duplication guard (Enhancement)	Rows de-duplicated by task and risk ID, so a mis-set node cannot inflate counts (found when counts came out 80× too high)
Langfuse observability (Core requirement)	One OpenTelemetry trace per request with orchestrator and agent spans, tags for route and prompt version, tokens, cost and latency
Human evaluation in Langfuse (Enhancement)	Annotation queue with two scores per trace: test_pass and pm_usefulness; all 12 baseline traces scored
Chat interface (Enhancement)	n8n chat trigger: PMs ask in plain language; a webhook stays available for automated test runs
Reports saved to Google Drive (Enhancement)	Every non-Clarifier answer becomes a formatted Google Doc in the Mira Outputs folder; the chat confirms it and links to the document, PDF download, Word download and folder
Scheduled weekly report (Enhancement)	Separate workflow every Friday 09:00 calls Mira as a sub-workflow for the current sprint and emails the report with the PDF attached
Test log with cost projection (Enhancement)	Workbook with per-test trace links, latency, tokens, cost, scores, findings and a monthly cost projection

Architecture



Orchestration pattern: Router. Requests vary, and each needs different data and a different prompt. A sequential pipeline would run every agent on every request, costing more and letting agents answer questions nobody asked. The router runs one agent. The trade-off, a single point of failure, is mitigated by recording the routing decision in every trace.

Output delivery (Enhancement). Clarifier answers stay in the chat. All other answers are converted to HTML, created as a formatted Google Doc in the Mira Outputs Drive folder, and the chat confirms the save with document, PDF, Word and folder links. The weekly workflow uses the same path and emails the report.

Submission contents

Everything below is in the GitHub repository. The Q1–Q4 documents are also submitted alongside the repository link.

File / folder	What it contains	Supports
workflows/Mira_Base.json	Main workflow: chat, webhook and sub-workflow triggers; Clean Input; Orchestrator; 4 Drive downloads; Route & Build Context; Switch; 6 agents; Langfuse tracing; Clarifier/document split; Google Doc creation; Drive reply	Q2 Core, Extended
workflows/ mira_weekly_report.json	Scheduled weekly workflow **(Enhancement)**: calls Mira for the current sprint and emails the report	Q2 Core: status report
README.md	Purpose, architecture, agents and tests, setup, assumptions, key findings	Guideline
docs/architecture.png, .html	Architecture diagram (static and interactive)	Q2 Design
docs/baseline_results.md	All 12 baseline tests: input, agent, result, answer contents; defects noted for v2	Q2 Baseline
docs/Mira_Test_Log.xlsx	Test Results (trace links, latency, tokens, cost, scores), Summary, Findings, Assumptions, Legend	Q2 Baseline, Cost; Q4
docs/prompts.md	All seven system prompts and temperatures	Q2 Design
code/ route_and_build_context.js	Routing rules and every computed fact	Q2 Grounding
code/build_langfuse_trace.js	OpenTelemetry trace builder: spans, tags, tokens	Q2 Observability
data/	The five input files from the brief	Reproducibility
screenshots/	Canvas; Mira answering T8 and T2; Drive reply and PDF; Langfuse traces, opened trace, annotation queue (12/12), dashboard; weekly workflow and email	Guideline
Q1 Ideation	Three use cases: pain point, solution, inputs/outputs, 3 metrics, knowledge base; prioritization	Q1 (15)
Q2 Build write-up	Architecture, pattern and tool rationale, agents, grounding, results, cost, evaluation, observations	Q2 (45)
Q3 Program charter	Vision, scope, success criteria, timeline, risks, stakeholders, decision-making, rollout	Q3 (25)
Q4 Reflection	Trace findings, improvement plan, evaluation-to-fine-tuning link, privacy, rollout to 10 PMs/TPMs	Q4 (15)

Where each Q2 criterion is evidenced

Q2 criterion (points)	Evidence
Design: architecture, orchestration, tool rationale (10)	Architecture diagram; Router justification; n8n vs Langflow rationale; prompts.md
Core capabilities end-to-end (12)	Planner (T1), Risk Assessor (T3, T7), Status Reporter (T5, T10) with upcoming milestones; documents in Drive; weekly report
Extended capability (8)	Milestone Alert System (T8, T11) and Stakeholder Update Generator (T12)
Observability connected (5)	Langfuse via OpenTelemetry; trace and dashboard screenshots
Baseline dataset tested with results (5)	baseline_results.md; Test Results tab; annotation queue 12/12 scored
Cost analysis and evaluation strategy (5)	$0.00046 per request, $0.78 per month projected; two human scores plus routing and refusal metrics

Running Mira (reviewer quick start)

Step	Action
1	Upload the four project files from data/ to a Google Drive folder; create a second folder named Mira Outputs
2	Import both workflows in n8n (Workflows → Import from file)
3	Add credentials: OpenAI (7 model nodes); Google Drive (4 downloads and the two Drive API nodes, with googleapis.com and www.googleapis.com allowed in the credential settings); Basic Auth for Langfuse (public key / secret key); Gmail for the weekly email
4	Set the Mira Outputs folder ID in the Google Doc and Document Reply nodes; reselect the main workflow in the weekly workflow's Call Mira node
5	Activate the main workflow, open the chat and ask T10. Expected: Done 5, In Progress 3, Blocked 1, To Do 16, saved to Drive with PDF and Word links

Key findings from building and testing

Finding	Root cause	Resolution
Planner wrote a full plan for a one-line chatbot request (T2)	No check that the request matched the loaded data	Orchestrator + rule-based routing + Clarifier
Status counts 80× too high	Drive downloads re-ran once per upstream item	Execute Once on downloads + de-duplication by ID
Brief says 3 tasks Done; the board has 5	Discrepancy in the brief	Mira reports the data
Langfuse rejected trace events	Legacy ingestion API closed for new projects	Switched to the OpenTelemetry endpoint
Status report lacked upcoming milestones	Status facts did not include the timeline	Added current phase, milestones and next phase (prompt v2)
Risk category occasionally reworded (T7)	LLM paraphrases fields it is told to copy	Planned: render risk fields in code
Re-runs changed wording, never facts	Facts are computed in code	Confirms the design principle
Google Doc creation blocked	Managed Google credential restricted HTTP Request use	Allowed googleapis.com domains in the credential
Chat showed raw API JSON after saving a document	The chat displays the last node's output field	Document Reply node writes a Drive confirmation message
Drive per-minute quota errors	Shared n8n Cloud Google app; several Drive calls per request	Spaced test runs; planned own OAuth app and caching
n8n workspace lost	Hosted workspace removed	Restored from exported JSON; workflows kept in GitHub

Known limitations

- One intent per request: "plan and top risks" returns one answer.
- Top-3 risk ranking is LLM judgment, because the register has no severity column.
- Token counts and per-step latency are estimated, not measured per LLM call.
- Document links open in the owner's Google account; others need access to the Mira Outputs folder.
- The Status Reporter sometimes omits the current-phase line (logged for v2).

Future enhancements

Ordered by value to PMs. The first item is the next build.

Enhancement	What changes	How in n8n
Choose how to receive each answer	After a report is ready, Mira asks in the chat: View in chat, Download PDF or Email it. Email asks for an address, validates it and sends the PDF	Chat node "Send and Wait for Response" with buttons; Switch on the choice; Drive download as PDF; Gmail with attachment
PDF served directly by n8n	Recipients download reports without access to the Drive folder	Download webhook → Drive download as PDF → Respond to Webhook with the file
PM approval before stakeholders	Weekly report goes to the PM with Approve/Reject buttons; only approved reports are sent on	Gmail "Send and wait for approval" in the weekly workflow
v2 prompt and grounding fixes	Fix the logged defects (T3, T7, T8, T9, T10, T11, T12)	Prompt edits; risk fields rendered in code; v1 vs v2 compared in Langfuse
Grounding validator	Every task and risk ID in an answer checked against the files; failures flagged and scored	Code node after the agents
Fine-tuned Risk Assessor	Model trained on corrected answers collected from traces	Langfuse dataset → OpenAI fine-tune → model swap; rerun T3/T7

Enhancement	What changes	How in n8n
Resource Allocation Advisor	Flags over- and under-allocated team members (Sarah Lin under near-term pressure; Ravi Kumar has capacity)	New agent; workload computed in code
LLM-as-a-judge evaluation	Every production trace scored for groundedness; humans review a weekly sample	Langfuse evaluators
Faster, quota-safe data loading	Files loaded once a day instead of on every request	Own Google OAuth app; cached data
Multi-intent requests	"Plan and top risks" answered together	Orchestrator returns a list; hierarchical pattern merges answers
Live task board	Reads Jira or Trello instead of CSV exports	Jira/Trello nodes in place of Drive downloads